

REPLICA METHOD FOR NEURAL NETWORKS

Bruno Fernández
MODELS MATEMÀTICS DE LA TECNOLOGIA

Supervised by Gabriel Mel

Motivation

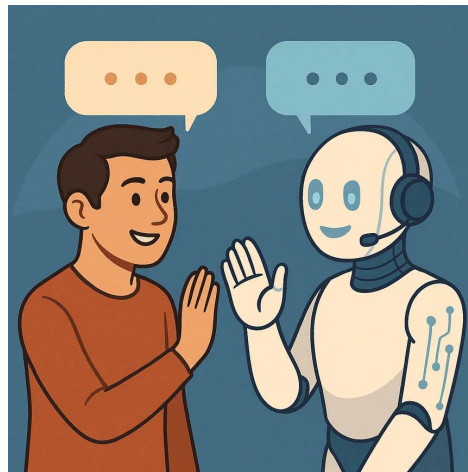
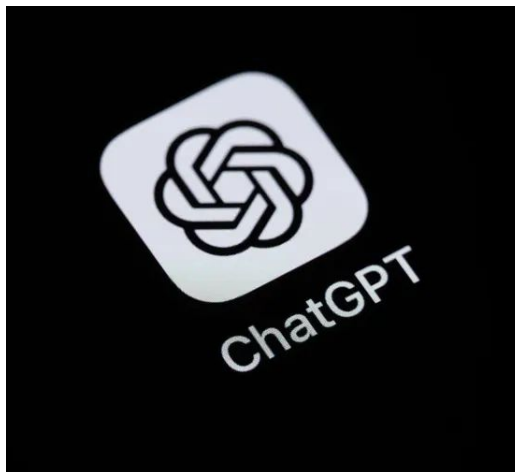


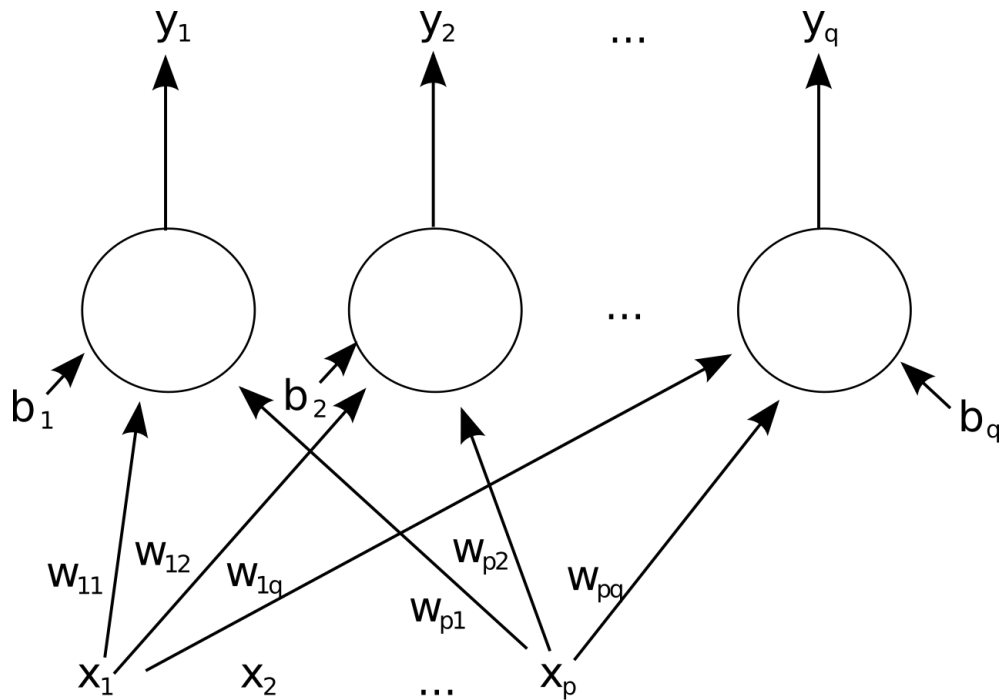


Table of contents

1. Introduction
2. Hopfield Network
3. Perceptron

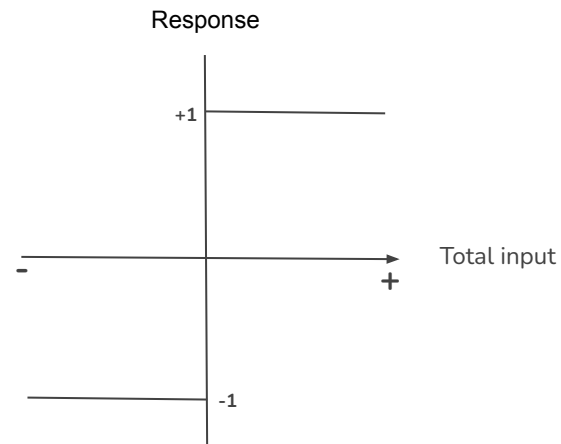
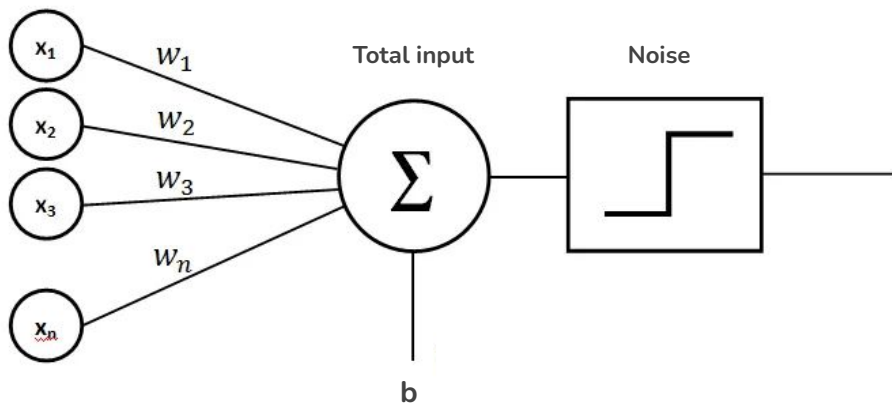


Neural Networks





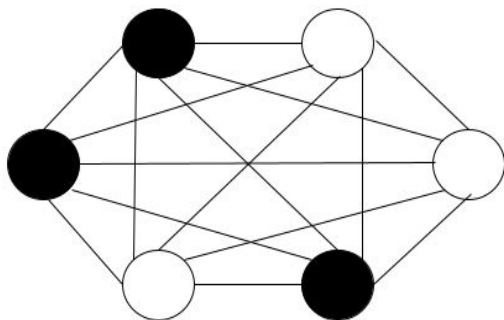
Neurons





Our Models

Hopfield network
Memory





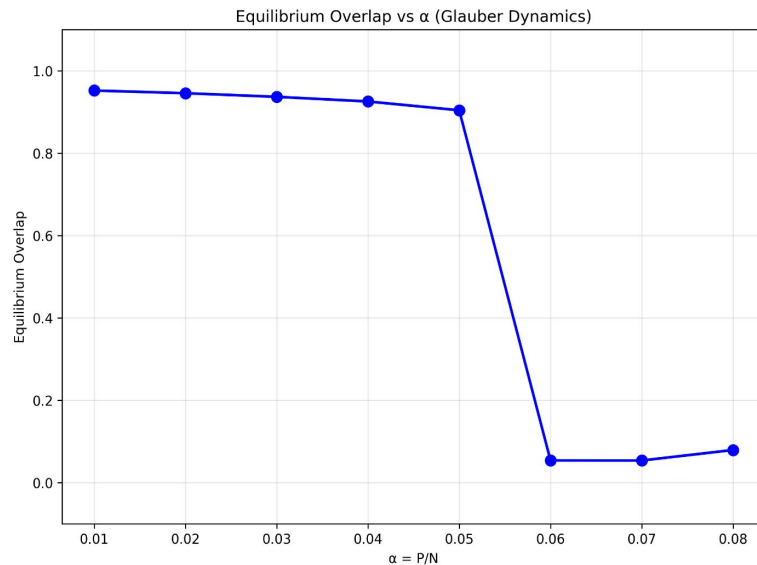
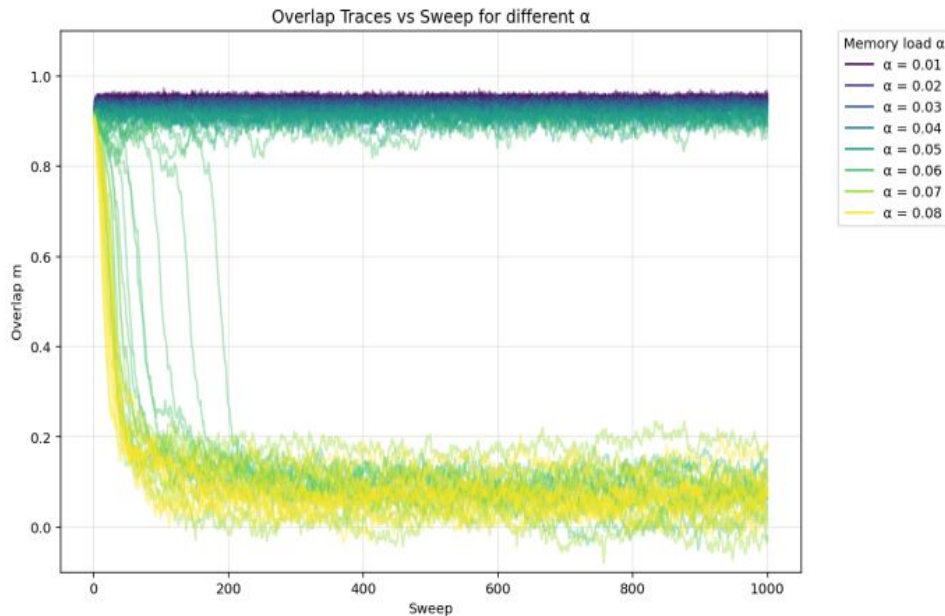
Simulations and results

$$\alpha = P/N$$

P = number of patterns

N = number of neurons

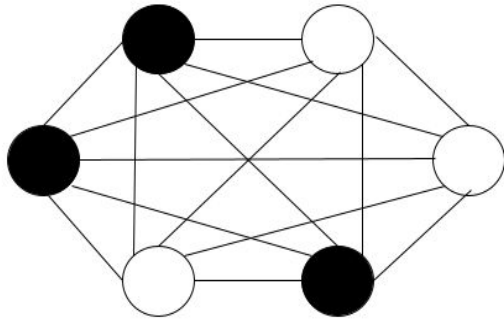
Glauber dynamics N = 4000



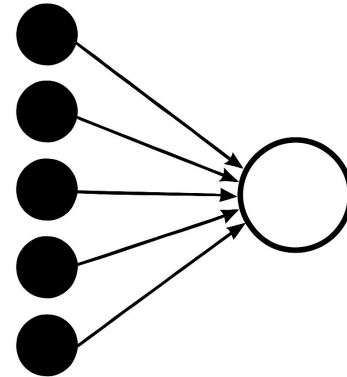


Our Models

Hopfield network
Memory



Perceptron
Decision making





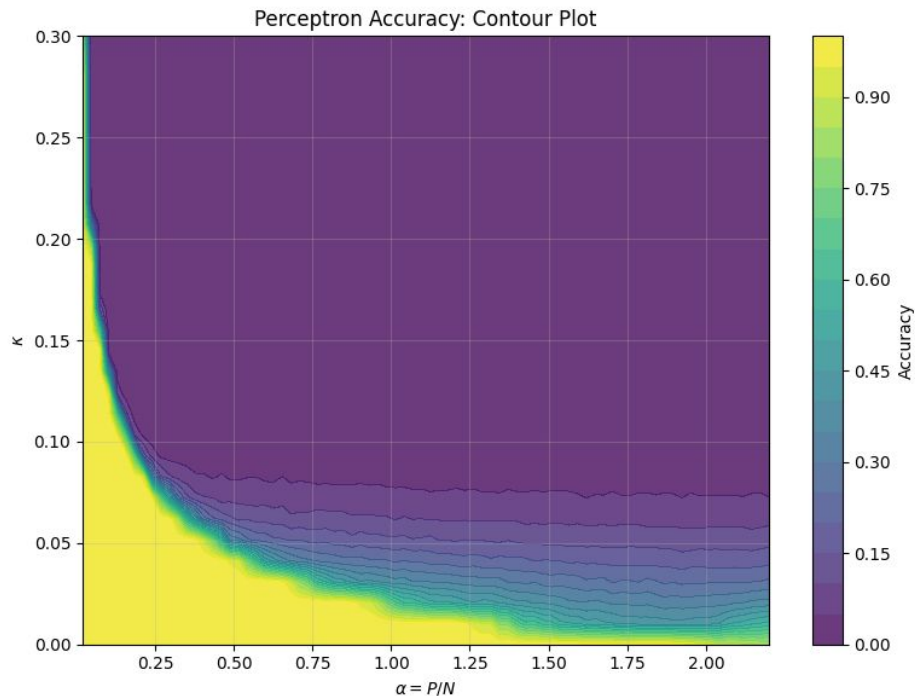
Simulations and results

Perceptron Accuracy

$$\alpha = P/N$$

P = number of patterns

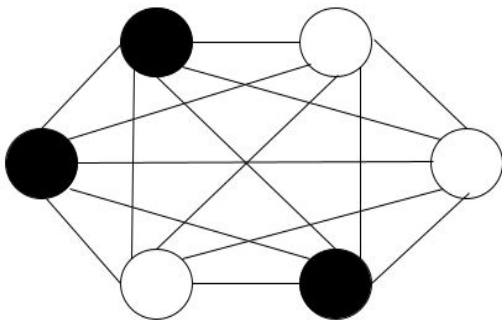
N = number of neurons



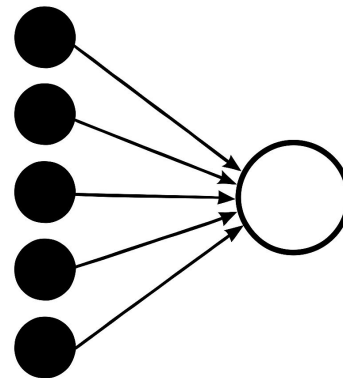


Goals

Hopfield network
Memory

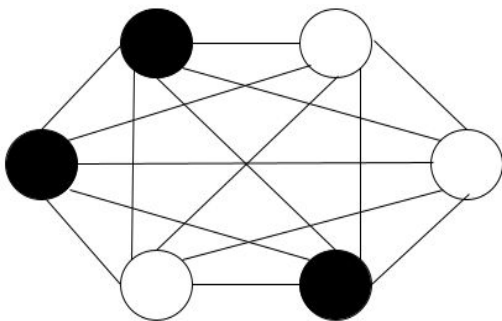


Perceptron
Decision making



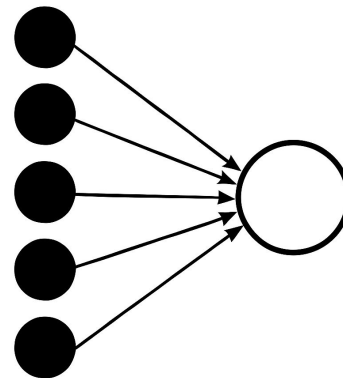
Goals

Hopfield network Memory



How many patterns can this neural network **remember**?

Perceptron Decision making



How many patterns can this neural network **classify**?



Replica Method

Free energy

$$f := \mathbb{E}_{x^{(1)} \dots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1, 1\}^N} e^{-\beta H(\sigma)}$$

$$\alpha = \frac{P}{N}$$

$$m$$

$$\beta = \frac{1}{T}$$



Replica Method

Free energy

$$f := \mathbb{E}_{x^{(1)} \dots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^N} e^{-\beta H(\sigma)}$$



Replica trick

$$\mathbb{E} \log f = \lim_{n \rightarrow 0} \frac{1}{n} \log \mathbb{E} f^n$$



Replica Method

Free energy

$$f := \mathbb{E}_{x^{(1)} \dots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1, 1\}^N} e^{-\beta H(\sigma)}$$



Replica trick

$$\mathbb{E} \log f = \lim_{n \rightarrow 0} \frac{1}{n} \log \mathbb{E} f^n$$



standard simplification steps



Replica Method

Free energy

$$f := \mathbb{E}_{x^{(1)} \dots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^N} e^{-\beta H(\sigma)}$$



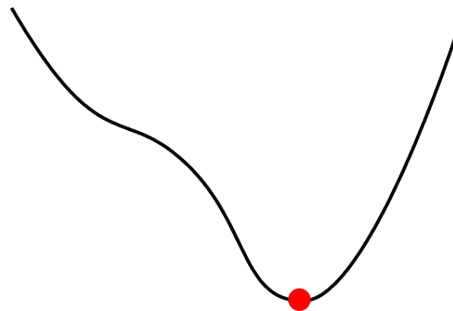
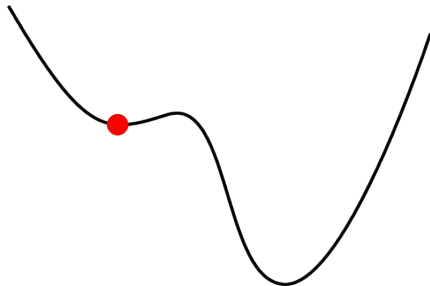
Replica trick

$$\mathbb{E} \log f = \lim_{n \rightarrow 0} \frac{1}{n} \log \mathbb{E} f^n$$



standard simplification steps

Potential Φ





HOPFIELD NETWORK

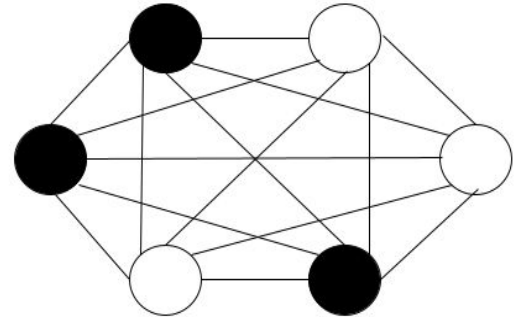


HOPFIELD NETWORK

N neurons, P patterns stored through a symmetric weight matrix J.

What is the **maximum memory capacity**?

We study the critical $\alpha = P/N$.





HOPFIELD NETWORK

What happens when there is **noise**?

The behaviour of the network follows **Boltzmann distribution**

$$P(\sigma) \propto e^{-\beta H(\sigma)} \quad \text{where } \beta = 1/T$$

Therefore, we study the free energy:

$$f := \mathbb{E}_{x^{(1)} \dots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1, 1\}^N} e^{-\beta H(\sigma)} \quad \xrightarrow[\text{Applying the **Replica Trick**}]{} \mathbb{E} \log f = \lim_{n \rightarrow 0} \frac{1}{n} \log \mathbb{E} f^n$$



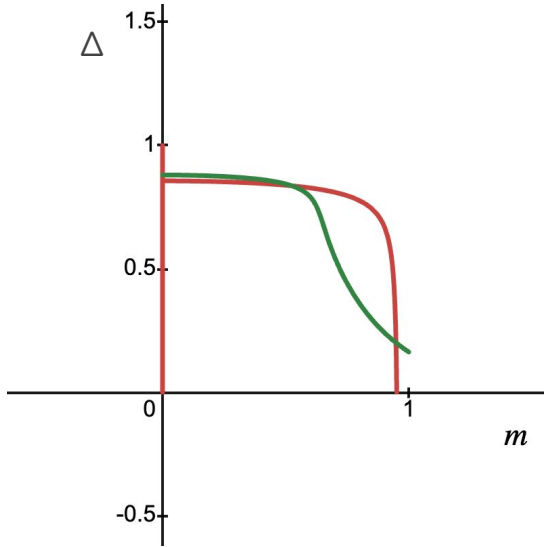
HOPFIELD NETWORK

m is **memory**
quality
 q is **replica overlap**

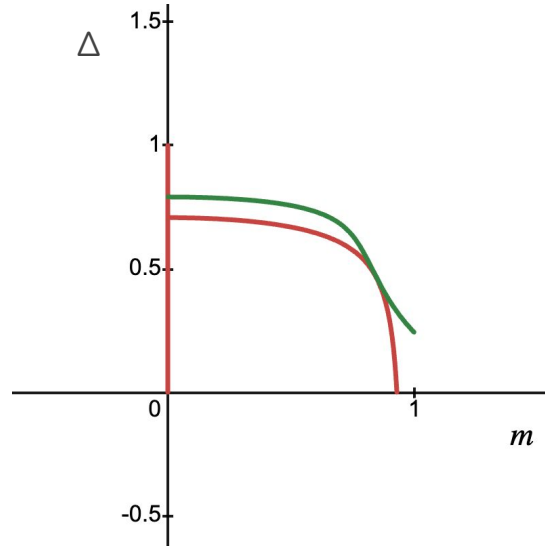
$$\Phi(m, q) \begin{cases} m = \mathbb{E}_z \left[\tanh \left(\beta \left(m + z \frac{\sqrt{\alpha q}}{(1 - \beta(1 - q))} \right) \right) \right] \\ q = \mathbb{E}_z \left[\tanh^2 \left(\beta \left(m + z \frac{\sqrt{\alpha q}}{(1 - \beta(1 - q))} \right) \right) \right] \end{cases}$$



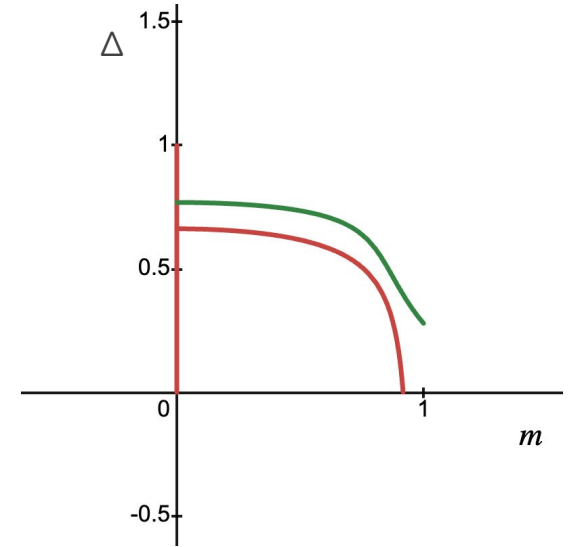
Free energy plots - Critical alpha value



$\alpha \approx 0$



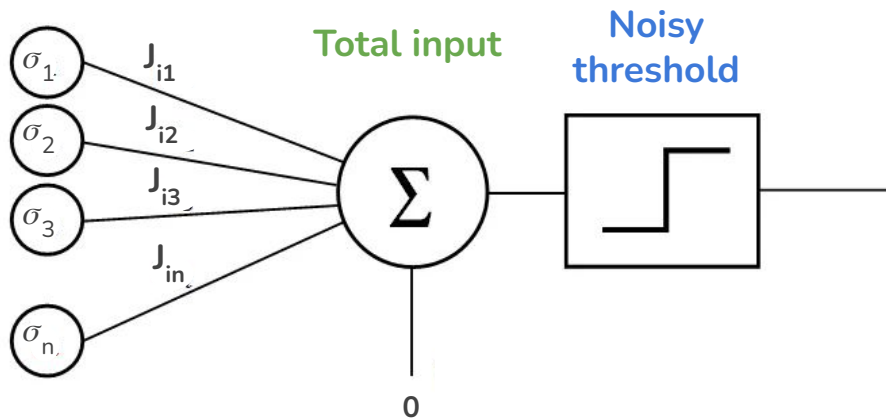
$0 < \alpha < \alpha^* \approx 0.06$



$\alpha > \alpha^*$

Testing the theory

Glauber Dynamics



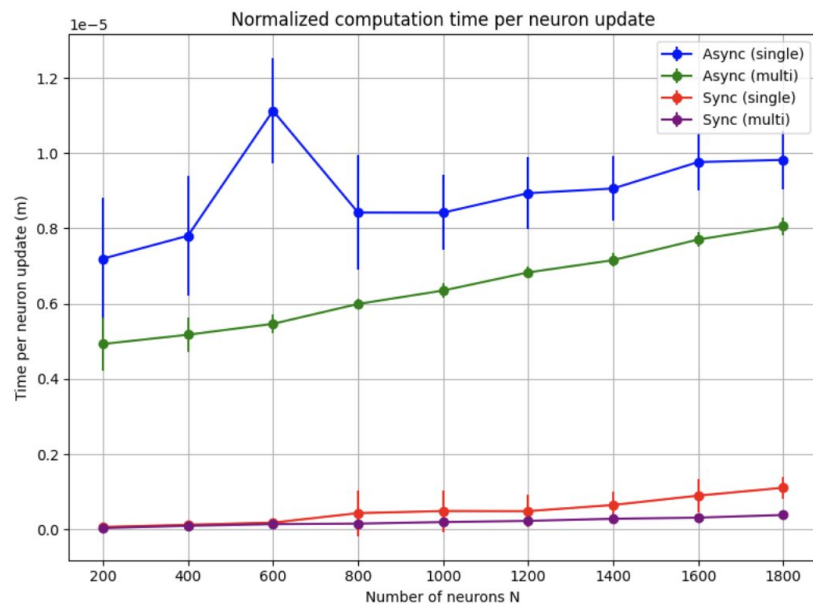
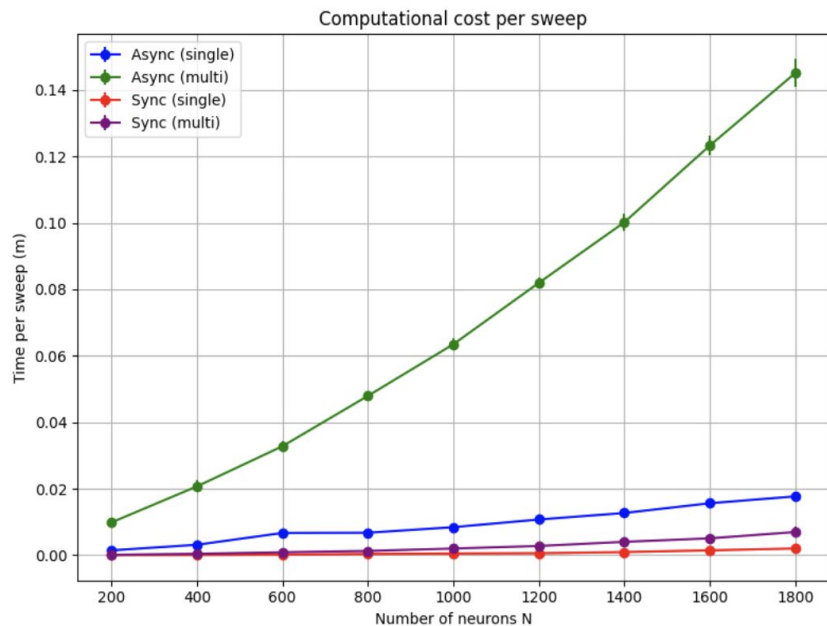
Total input

$$h_i = \sum_{j \neq i} J_{ij} \sigma_j$$

Noisy threshold

$$\sigma_i \rightarrow \text{sgn} \left(\sum_{j \neq i} J_{ij} \sigma_j \right)$$

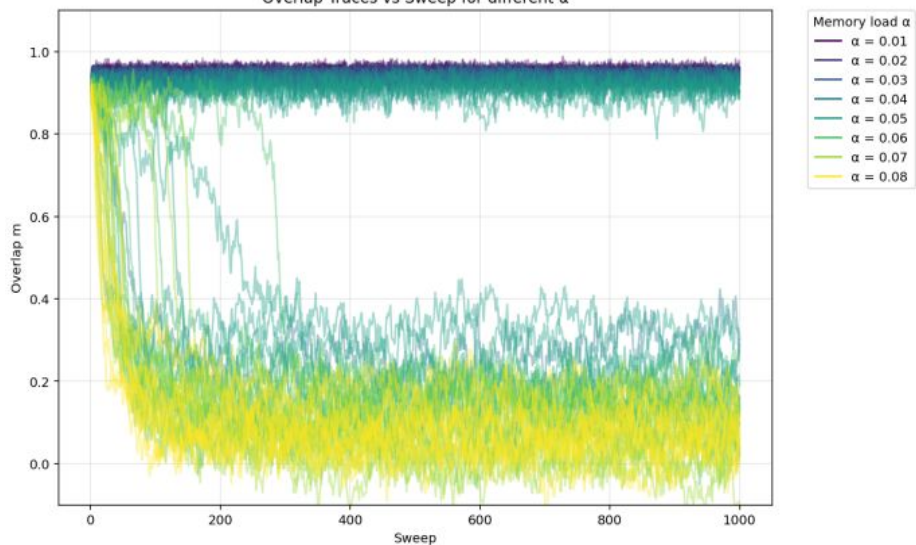
Computational cost (Sync/Async)



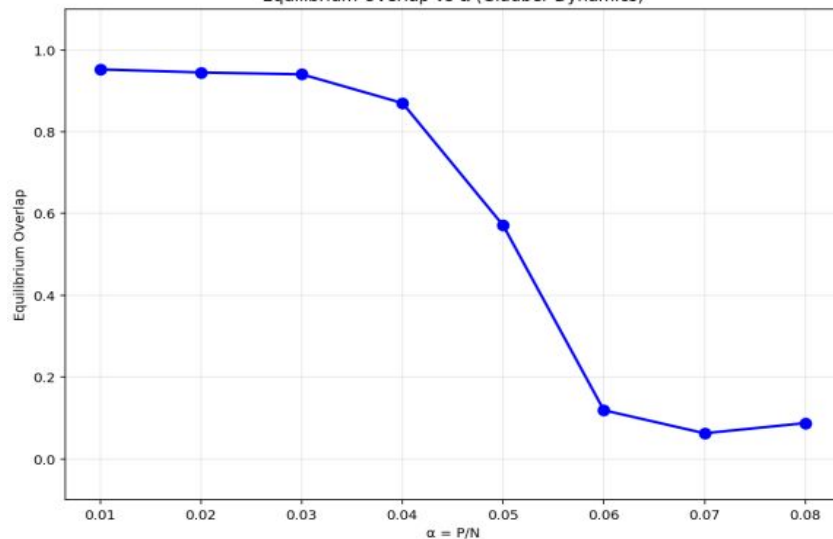


Glauber panels $N = 1000$

Overlap Traces vs Sweep for different α

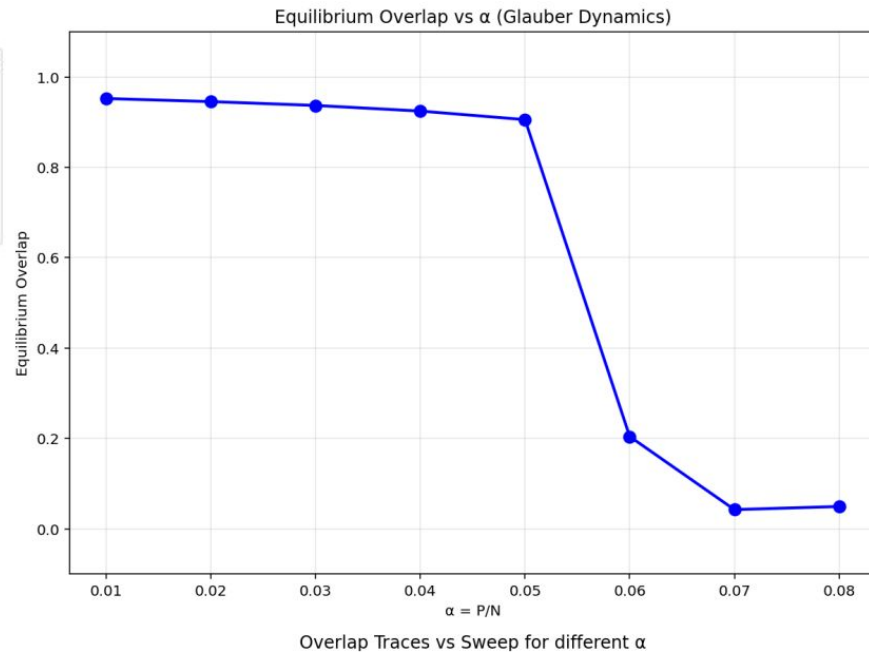
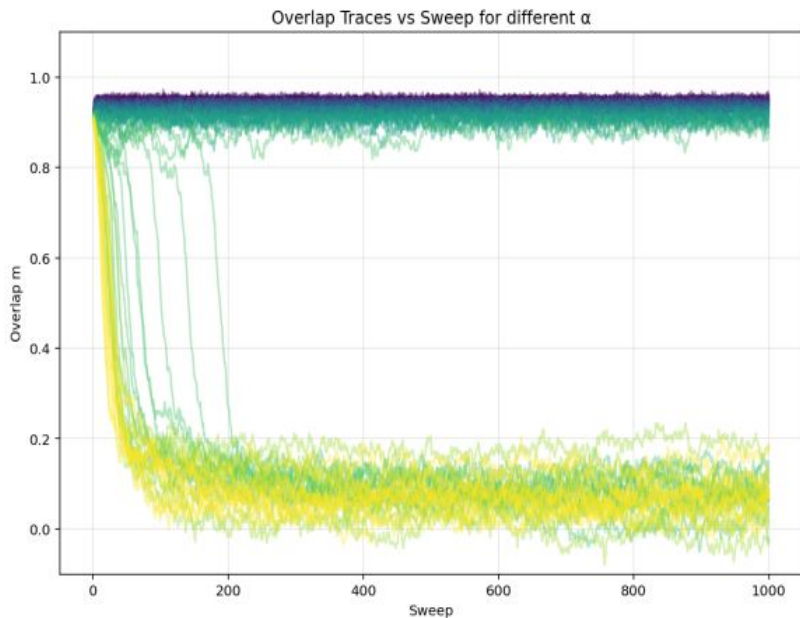


Equilibrium Overlap vs α (Glauber Dynamics)



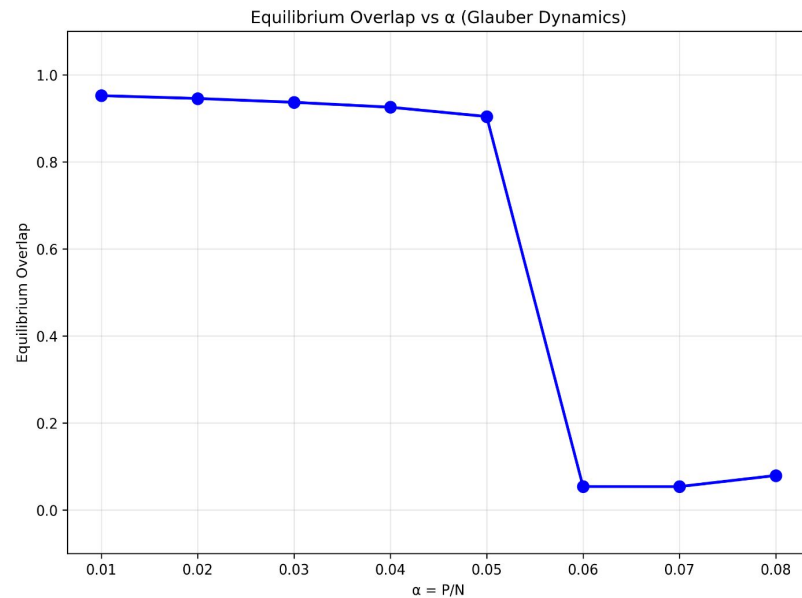
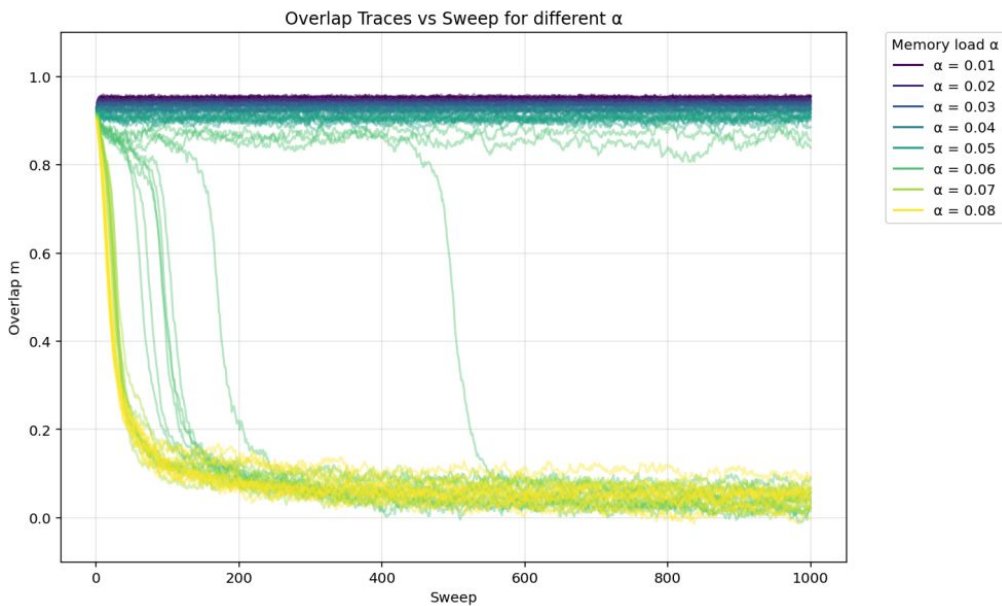


Glauber panels $N = 4000$



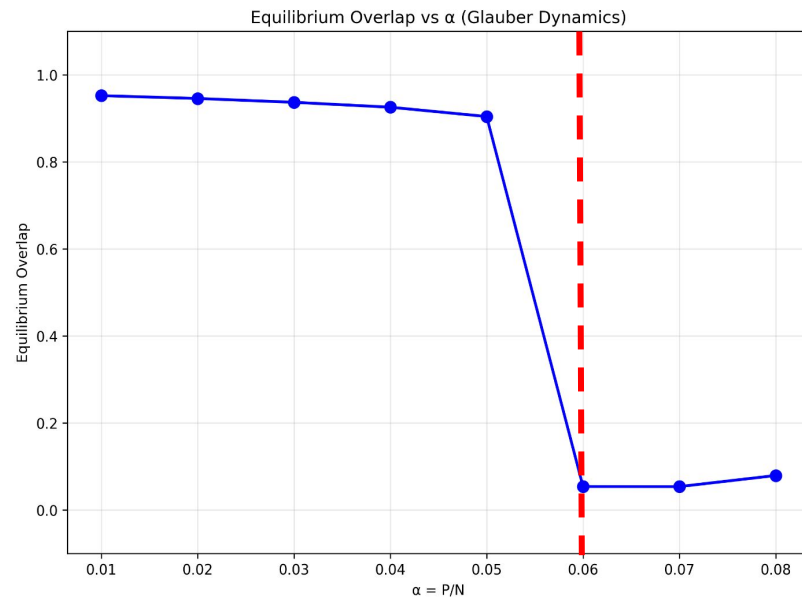
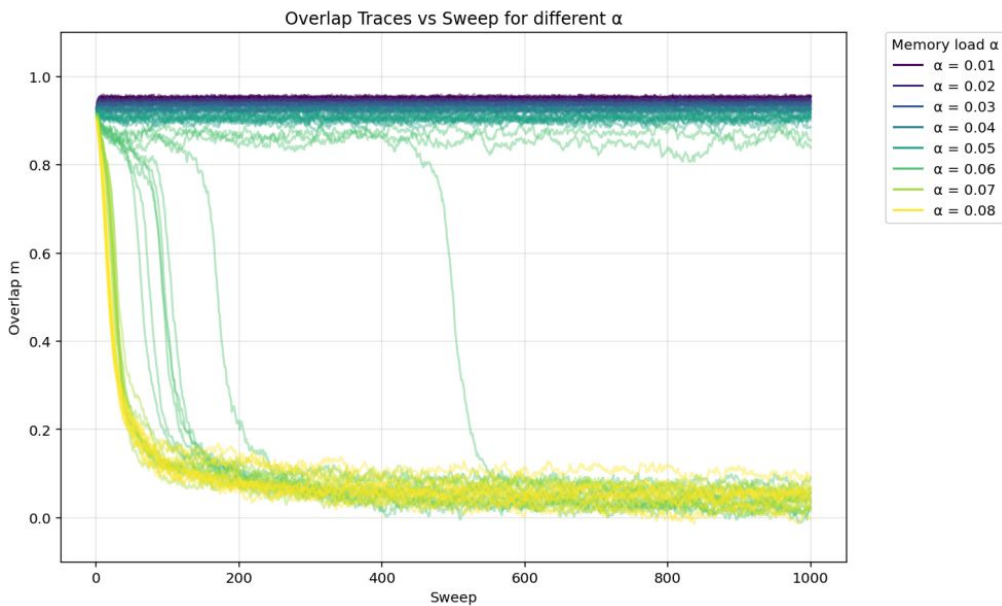


Glauber panels $N = 16000$





Glauber panels $N = 16000$

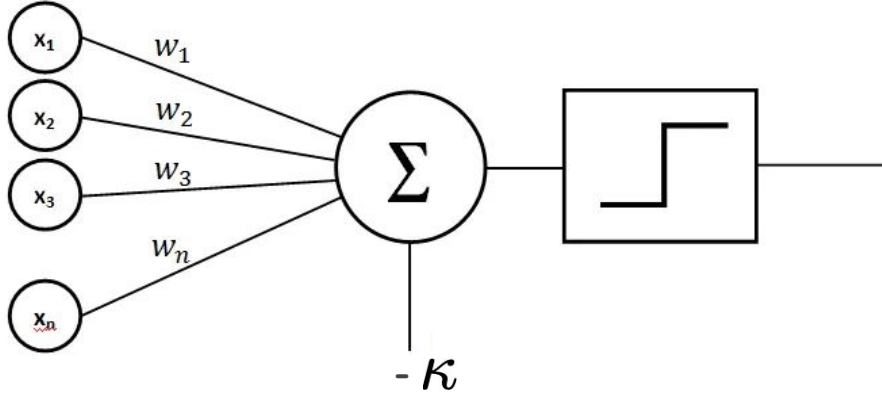




PERCEPTRON

PERCEPTRON

How many patterns can this neural network **correctly classify**?



$$\sigma_i = \Theta \left(\frac{w \cdot x^{(\mu)}}{\sqrt{N}} - K \right)$$



PERCEPTRON

Integration among the volume that correctly classifies all patterns.

$$\mathbb{E}_x \frac{1}{N} \log \int_{S^{N-1}} dw \left[\prod_{\mu=1}^P \Theta \left(\frac{w \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right) \right]$$

$$\mathbb{E} \log f = \lim_{n \rightarrow 0} \frac{1}{n} \log \mathbb{E} f^n$$



Applying the **Replica Trick**

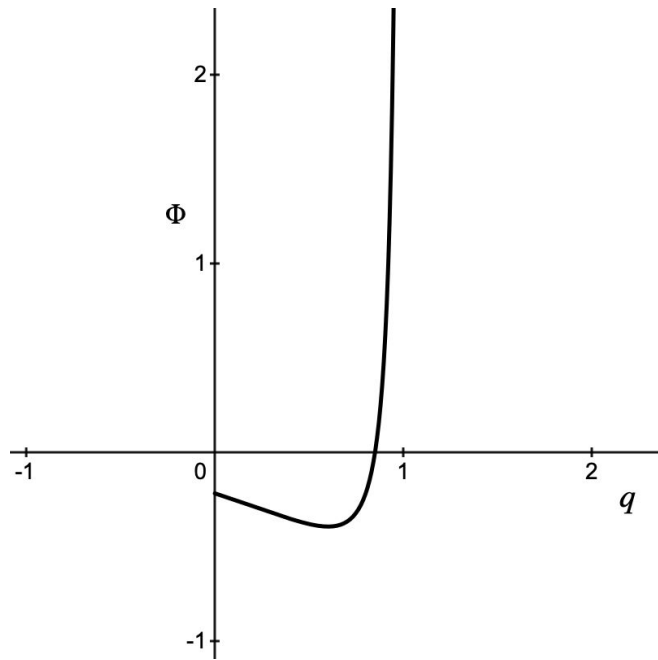


PERCEPTRON

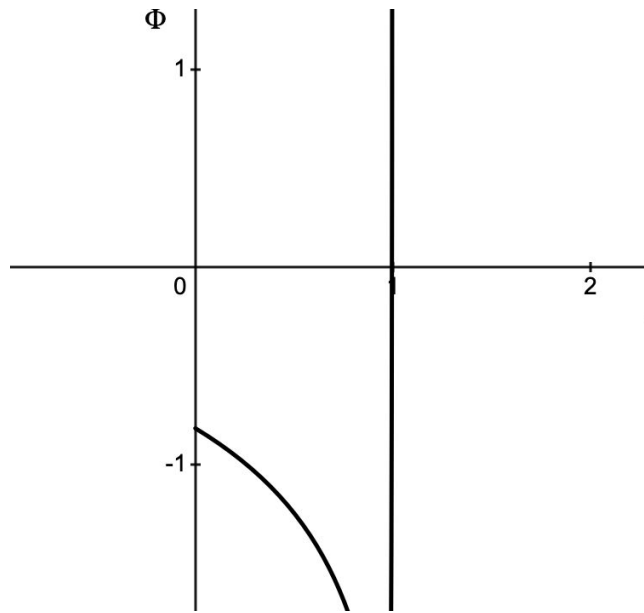
$$\log V = \min_q \Phi(q)$$



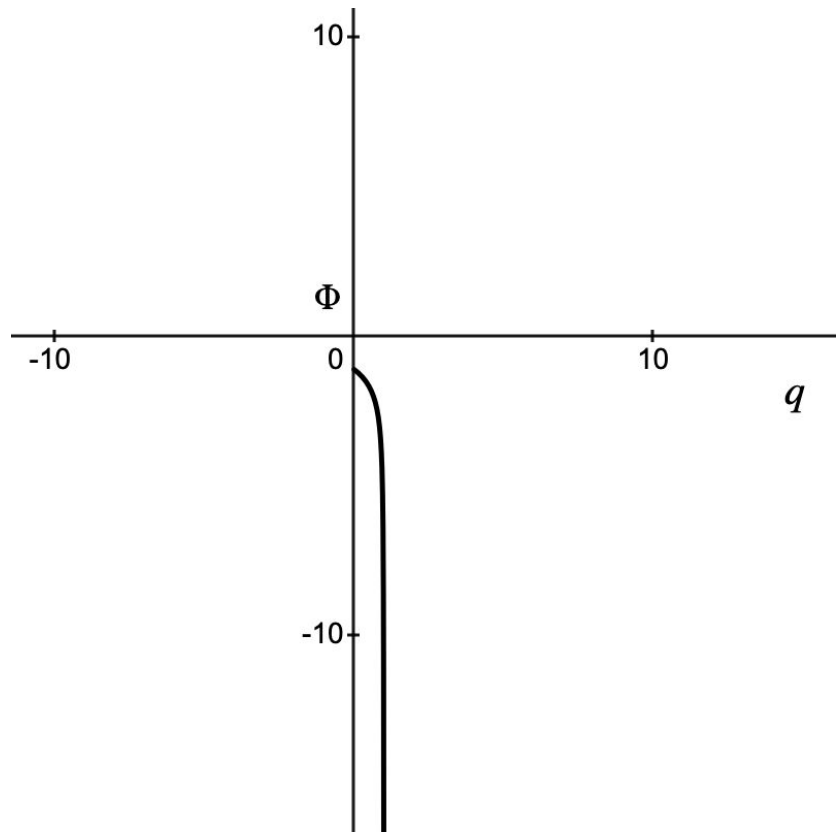
$$\Phi(q) = \alpha \mathbb{E}_z \left[\log \mathbb{P} \left(u \geq \frac{\kappa - \sqrt{q} z}{\sqrt{1-q}} \right) \right] + \frac{1}{2} \frac{1}{1-q} + \frac{1}{2} \log(1-q)$$



$\alpha \approx 0$



$0 < \alpha < \alpha^*$



$$\alpha > \alpha^*$$

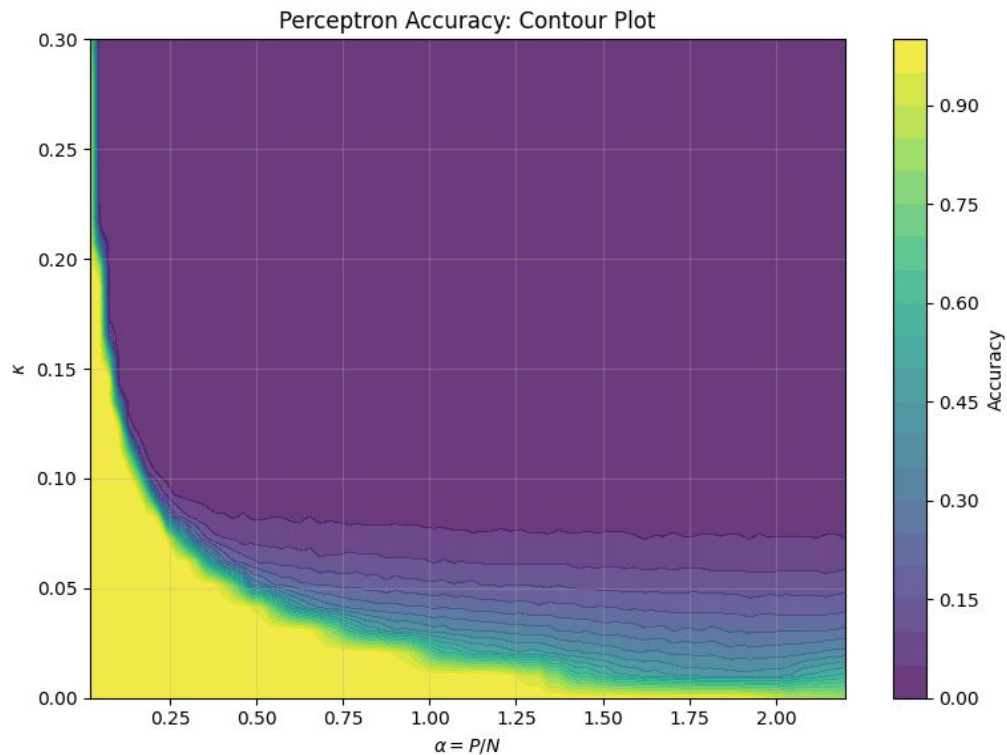


CRITICAL ALPHA VALUE

$$\Phi(q) = \alpha \mathbb{E}_z \left[\log \mathbb{P} \left(u \geq \frac{\kappa - \sqrt{q} z}{\sqrt{1-q}} \right) \right] + \frac{1}{2} \frac{1}{1-q} + \frac{1}{2} \log(1-q)$$

$$\alpha_c^{-1} = \int_{-\infty}^{\kappa} Dz (\kappa - z)^2$$

Perceptron Accuracy



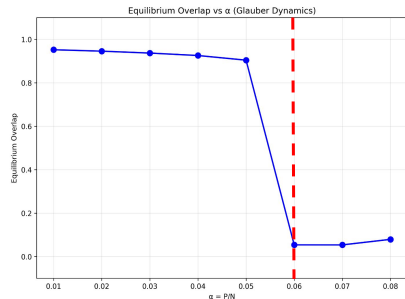
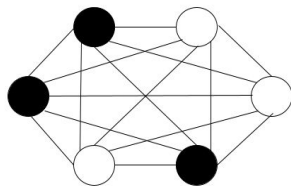
$$\alpha_c^{-1} = \int_{-\infty}^{\kappa} Dz (\kappa - z)^2$$



Summary / Conclusions

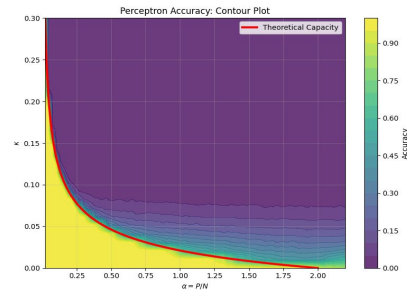
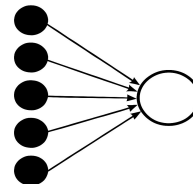
- Hopfield Network

How many patterns can this neural network **remember**?

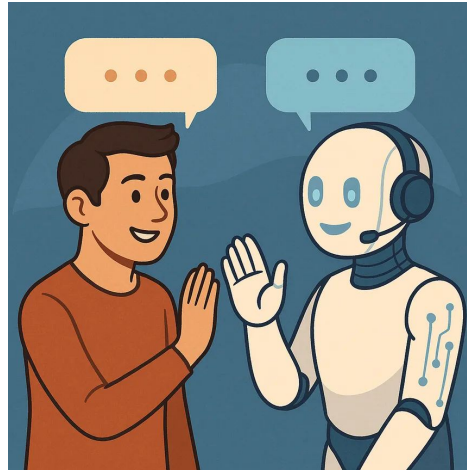


- Perceptron

How many patterns can this neural network **classify**?



Future Work



The end.